Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 10: Waveguides 2

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1st Midterm

- Closed-book.
- One page (front and back) of handwritten notes is allowed.
- Notes can contain equations only.
- All exam problems will be from in-class examples or homework problems.
- Coverage: Lecture 1-8 and Homework 1-4.

What is the boundary condition for this Laplace Equation (for Ez and Bz)?

$$\bar{E}_z(\vec{x}, t) = XY e^{i(kz - \omega t)}$$

$$\begin{cases} (\nabla_t^2 - k^2 + \mu\varepsilon\omega^2) XY = 0 \\ E_z = 0 \quad \text{on} \quad S \Rightarrow XY = 0 \quad \text{on} \quad S. \end{cases}$$

B_z

$$\begin{cases} (\nabla_t^2 - k^2 + \mu\varepsilon\omega^2) XY = 0 \\ \frac{\partial B_z}{\partial n} = 0 \quad \text{on} \quad S \Rightarrow \frac{\partial (XY)}{\partial n} = 0 \quad \text{on} \quad S \end{cases}$$

Can we get transverse component from parallel component?

$$\vec{E}(\vec{x},t) = \vec{E}_z + \vec{E}_t \qquad \vec{E}_z = \hat{z}\,E_z \qquad \vec{E}_t = (\hat{z}\times\vec{E})\times\hat{z}$$

① $\nabla\cdot\vec{E}=0$

② $\nabla\cdot\vec{B}=0$

$$\left(\nabla_t + \hat{z}\frac{\partial}{\partial z}\right)\cdot\left(\vec{E}_z + \vec{E}_t\right) = 0$$

$$\nabla_t\cdot\vec{E}_z + \nabla_t\cdot\vec{E}_t + \hat{z}\cdot\frac{\partial\vec{E}_z}{\partial z} + \hat{z}\cdot\frac{\partial\vec{E}_t}{\partial z} = 0$$

($\nabla_t\cdot\vec{E}_z = 0$, $\hat{z}\cdot\frac{\partial\vec{E}_t}{\partial z} = 0$)

$$\Rightarrow \quad \nabla_t\cdot\vec{E}_t = -\frac{\partial E_z}{\partial z} \qquad \Rightarrow \nabla_t\cdot\vec{B}_t = -\frac{\partial B_z}{\partial z}$$

Express Bt and Et with Bz and Ez

$$\nabla \times \vec{E} = i\omega \vec{B} \Rightarrow (\nabla_t + \frac{\partial}{\partial z}\hat{z}) \times (\vec{E}_t + E_z) = i\omega (B_t + B_z)$$

$$\nabla_t \times \vec{E}_t + \nabla_t \times E_z + \hat{z} \times \frac{\partial \vec{E}_t}{\partial z} + \hat{z} \times \frac{\partial E_z}{\partial z} = i\omega (B_t + B_z)$$

$\frac{\partial E_z}{\partial z} = 0$

$$\begin{cases} \nabla_t \times \vec{E}_t = i\omega B_z & ③ \\ \nabla_t \times E_z + \hat{z} \times \frac{\partial \vec{E}_t}{\partial z} = i\omega B_t \end{cases}$$

$\nabla_t \times (E_z \hat{z}) = (\nabla_t E_z) \times \hat{z}$

$$\hat{z} \times \left[(\nabla_t E_z) \times \hat{z} + \hat{z} \times \frac{\partial \vec{E}_t}{\partial z} = i\omega B_t \right]$$

$$\nabla_t E_z - \frac{\partial \vec{E}_t}{\partial z} = i\omega \hat{z} \times \vec{B}_t \quad ④$$

$$\Rightarrow \frac{\partial \vec{E}_t}{\partial z} + i\omega \hat{z} \times \vec{B}_t = \nabla_t E_z$$

Express Bt and Et with Bz and Ez

$$\nabla \times \vec{B} = -i\mu\varepsilon\omega\vec{E}$$

$$\begin{cases} \nabla_t \cdot \vec{E}_t = -\dfrac{\partial E_z}{\partial t} & ① \\ \nabla_t \cdot \vec{B}_t = -\dfrac{\partial B_z}{\partial t} & ② \\ \hat{z} \cdot (\nabla_t \times \vec{E}_t) = i\omega B_z & ③ \\ \dfrac{\partial \vec{E}_t}{\partial z} + i\omega \hat{z} \times \vec{B}_t = \nabla_t E_z & ④ \\ \hat{z} \cdot (\nabla_t \times \vec{B}_t) = -i\mu\varepsilon\omega E_z & ⑤ \\ \dfrac{\partial \vec{B}_t}{\partial t} - i\mu\varepsilon\omega \hat{z} \times \vec{E}_t = \nabla_t B_z & ⑥ \end{cases}$$

Express Bt and Et with Bz and Ez

$$\textcircled{4} \Rightarrow \quad ik\vec{E}_t + i\omega \hat{z} \times \vec{B}_t = \nabla_t E_z$$

$$\textcircled{6} \Rightarrow \quad ik\vec{B}_t - i\mu\varepsilon\omega \hat{z} \times \vec{E}_t = \nabla_t B_z$$

$$\Rightarrow \quad \vec{B}_t = \frac{\mu\varepsilon\omega}{k} \hat{z} \times \vec{E}_t - \frac{i}{k} \nabla_t B_z$$

$$\Rightarrow \begin{cases} \vec{E}_t = \dfrac{i}{\mu\varepsilon\omega^2 - k^2} \left[k\nabla_t E_z - \omega \hat{z} \times \nabla_t B_z \right] \\ \vec{B}_t = \dfrac{i}{\mu\varepsilon\omega^2 - k^2} \left[k\nabla_t B_z + \mu\varepsilon\omega \hat{z} \times \nabla_t E_z \right] \end{cases}$$

Summary: Steps to solve the wave eq

1. Solve 2D Laplace Eq for Ez or Bz
2. Use Ez or Bz to get Et and Bt

Different modes of the solution

- Bz=0 and Ez!=0
- Bz!=0 and Ez=0
- Bz=0 and Ez=0

Bz=0 and Ez!=0

$$\begin{cases} \vec{E}_t = \dfrac{i}{\mu\varepsilon\omega^2 - k^2}\left[k\nabla_t E_z - \omega \hat{z} \times \nabla_t B_z \right] \\ \vec{B}_t = \dfrac{i}{\mu\varepsilon\omega^2 - k^2}\left[k\nabla_t B_z + \mu\varepsilon\omega \hat{z} \times \nabla_t E_z \right] \end{cases}$$

Transverse Magnetic mode TM mode

$$E_z(\vec{x}, t) = XY e^{i(kz - \omega t)}$$

$$\begin{cases} (\nabla_t^2 - k^2 + \mu\varepsilon\omega^2) XY = 0 \\ E_z = 0 \text{ on } S \Rightarrow XY = 0 \text{ on } S. \end{cases}$$

Bz!=0 and Ez=0

Transverse Electric mode (TE)

$$\begin{cases} \vec{E}_t = \dfrac{i}{\mu\varepsilon\omega^2 - k^2}\left[k\nabla_t \bar{E}_z - \omega \vec{z} \times \nabla_t B_z\right] \\ \vec{B}_t = \dfrac{i}{\mu\varepsilon\omega^2 - k^2}\left[k\nabla_t B_z + \mu\varepsilon\omega \vec{z} \times \nabla_t \bar{E}_z\right] \end{cases}$$

$$\begin{cases} (\nabla_t^2 - k^2 + \mu\varepsilon\omega^2) X Y = 0 \\ ? \quad \text{on} \quad S \end{cases} \qquad B_z = X Y e^{i(kz - \omega t)}$$

Boundary condition for Bz

$$\vec{B}_t = \frac{i}{\mu\varepsilon\omega^2 - k^2} k \nabla_t B_z$$

BC: $\vec{B}_\perp = 0$ $\qquad \vec{n}\cdot\vec{B} = 0$

$$\vec{n}\cdot(\vec{B}_t + \vec{B}_z) = 0 \Rightarrow \vec{n}\cdot\vec{B}_t = 0$$

$$\Rightarrow \vec{n}\cdot\nabla_t B_z = 0$$

$$\Rightarrow \frac{\partial B_z}{\partial n} = 0$$

$$\begin{cases} (\nabla_t^2 - k^2 + \mu\varepsilon\omega^2) X Y = 0 \\ \frac{\partial B_z}{\partial n} = 0 \text{ on } S \end{cases}$$

Bz=0 and Ez=0

$$① \Rightarrow \nabla_t \cdot \vec{b}_t = -\frac{\partial b_z}{\partial z} = 0$$

$$⑤ \Rightarrow \hat{z} \cdot (\nabla_t \times \vec{b}_t) = i\omega \beta_z = 0$$

$$\Rightarrow \nabla_t \times \vec{b}_t = 0$$

$$\Rightarrow \nabla^2 \vec{b}_t = 0 \Rightarrow (\nabla_t^2 - k^2 + \mu\varepsilon\omega^2)\, \vec{b}_t = 0$$

$$k^2 = \mu\varepsilon\omega^2 \qquad k = \omega\sqrt{\mu\varepsilon}$$

Bz=0 and Ez=0

⑥ $\Rightarrow \quad ik\vec{B}_t - i\omega\mu\varepsilon\,\hat{z}\times\vec{E}_t = \nabla_t B_z = 0$

$$\vec{B}_t = \frac{\omega\mu\varepsilon}{k}\,\hat{z}\times\vec{E}_t$$

$$\vec{B}_t \perp \vec{E}_t \quad \Rightarrow \quad \vec{B} \perp \vec{E}$$

TEM mode

Transverse Electric Magnetic

Bz=0 and Ez=0

$$\vec{b}_t = -\nabla\varphi$$

$$\begin{cases} \nabla_t \cdot \vec{b}_t = 0 \\ \nabla_t \times \vec{b}_t = 0 \end{cases}$$

Perfect conductor

$$\Rightarrow \quad \begin{cases} \nabla_t^2 \varphi = 0 \end{cases}$$

Perfect conductor

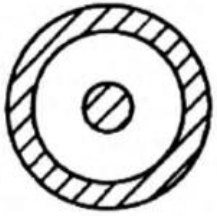
$$\Downarrow$$

$$\varphi|_S = \text{Const}$$

$$\varphi = \text{Const} \quad . \quad \vec{b}_t = 0$$

Bz=0 and Ez=0

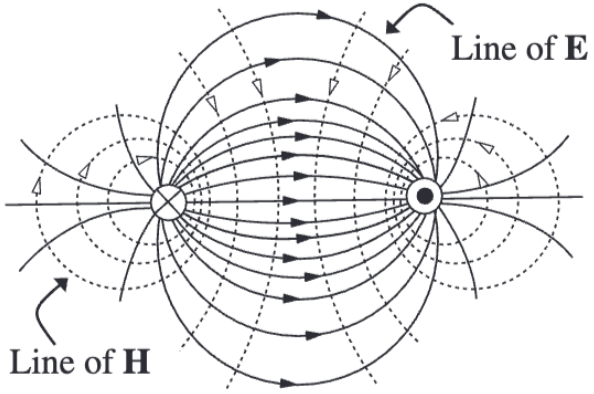
- Transmission lines



Coaxial line



Parallel wires



Waveguides

- TM $(B_z = 0)$
- Express H_t and E_t with Ez

$$\vec{E}_t = \frac{ik}{\mu\varepsilon\omega^2 - k^2} \nabla_t E_z \qquad \gamma^2 = \mu\varepsilon\omega^2 - k^2$$

$$= \frac{ik}{\gamma^2} \nabla_t E_z \qquad \vec{H}_t = \frac{i}{\gamma^2} \varepsilon\omega \hat{z} \times \nabla_t E_z$$

$$= \frac{1}{Z_{TM}} \hat{z} \times \vec{E}_t \qquad Z_m = \frac{k}{\varepsilon\omega}$$

- 2D wave Eq

$$\begin{cases} \nabla_t^2 E_z + \gamma^2 E_z = 0 \\ E_z = 0 \quad \text{on} \quad S. \end{cases}$$

Waveguides

- TE $(\bar{B}_z = 0)$

2D $\begin{cases} \nabla_t^2 B_z + \gamma^2 B_z = 0 \\ \dfrac{\partial B_z}{\partial n} = 0 \quad \text{on } S \end{cases}$

$$\vec{E}_t = \frac{i}{\gamma^2}(-\omega \vec{z} \times \nabla_t B_z)$$

$$\vec{H}_t = \frac{ik}{\omega \mu \gamma^2} \nabla_t B_z$$

$$\vec{H}_t = \frac{1}{Z_{TE}} \vec{z} \times \vec{E}_t \qquad Z_{TE} = \frac{\omega \mu}{k}$$